

Christoffer G. Alexandersen

MATHEMATICAL MODELING OF NEURODEGENERATIVE DISEASE · NEUROIMAGING AND MULTIMODAL DATA ANALYSIS · BAYESIAN INFERENCE AND PARAMETER ESTIMATION

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Academic Employment

Yale University POSTDOCTORAL RESEARCHER

July 2026 – Present, New Haven, CT, USA

Faculty of Arts and Sciences
Department of Psychology
Complex Systems Lab
Principal Investigator: Prof. Dani S. Bassett

University of Pennsylvania POSTDOCTORAL RESEARCHER

June 2024 – June 2026, Philadelphia, PA, USA

School of Engineering and Applied Science
Department of Bioengineering
Complex Systems Lab
Principal Investigator: Prof. Dani S. Bassett

Education

University of Oxford PH.D. MATHEMATICS

Oct. 2020 - Sept. 2024, Oxford, UK

- Thesis: Mathematical modeling of neuronal dynamics during disease
- Advisors: Prof. Alain Goriely and Prof. Christian Bick

Norwegian University of Science and Technology 5-YEAR MSc. COMPUTATIONAL BIOLOGY

Sept. 2015 - July 2020, Trondheim, Norway

- Thesis: Acknowledging the uncertainty of enzyme kinetic parameters in constraint-based metabolic modeling
- Advisor: Prof. Eivind Almaas

Publications

1. **Alexandersen CG**, Brennan GS, Brynildsen JK, Henderson MX, Iturria-Medina Y, Bassett DS. Network Models of Neurodegeneration: Bridging Neuronal Dynamics and Disease Progression. *IEEE Reviews in Biomedical Engineering*. 19:140–158. doi:10.1109/RBME.2025.3643310. 2026.
2. **Alexandersen CG**, Bassett D, Goriely A, Chaggar P. Neuronal activity and amyloid- β promote tau seeding in the entorhinal cortex in Alzheimer's disease. *Brain*, <https://doi.org/10.1093/brain/awaf374>. 2025
3. Cabrera-Álvarez J, del Cerro León A, Carvajal BP, Carrasco-Gómez M, **Alexandersen CG**, Bruña R, Maestú F, Susi G. The fluctuations of alpha power: bimodalities, connectivity, and neural mass models. *Imaging Neuroscience*. 3 IMAG.a.64. doi:10.1162/IMAG.a.64. 2025.
4. Borgquist J, **Alexandersen CG**. HeMiTo-dynamics: a characterization of mammalian prion toxicity using non-dimensionalization, linear stability and perturbation analyses. *Mathematical Medicine and Biology*. Volume 42, Issue 2, Pages 159–175. doi:10.1093/imammb/dqae024. 2025
5. **Alexandersen CG**, Goriely A, Bick C. Neuronal activity induces symmetry breaking in neurodegenerative disease spreading. *Journal of Mathematical Biology*. 89, 3. doi:10.1007/s00285-024-02103-x. 2024
6. **Alexandersen CG**, Duprat C, Ezzati A, Houzelstein P, Ledoux A, Liu Y, Saghir S, Destexhe A, Tesler F, Depannemaecker D. A mean-field to capture asynchronous irregular dynamics of conductance-based networks of adaptive quadratic integrate-and-fire neuron models. *Neural Computation*. 36 (7): 1433–1448. doi:10.1162/neco_a_01670. 2024
7. **Alexandersen CG**, Douw L, Zimmermann MLM, Bick C, Goriely A. Functional connectotomy of a whole-brain model reveals tumor-induced alterations to neuronal dynamics in glioma patients. *Network Neuroscience*. 9 (1): 280–302. doi:10.1162/netn_a_00426. 2024
8. **Alexandersen CG**, de Haan W, Bick C, Goriely A. A multi-scale model explains oscillatory slowing and neuronal hyperactivity in Alzheimer's disease. *Journal of the Royal Society Interface*. 20: 20220607. doi:10.1098/rsif.2022.0607. 2023

Preprints

1. Mohapatra S, **Alexandersen CG**, Fotiadis P, Kelz MB, Detre JA, Pasqualetti F, Bassett DS. QUIET: Quantifying Underutilized Influential Edges for Targeted Synchronization. *arXiv*. doi: <https://doi.org/10.48550/arXiv.2606.11091> . 2026.
2. **Alexandersen CG**, Brynildsen JK, Prigent A, Tamborrino M, Mantziou A, Kurgat K, Henderson MX, Bassett DS. Rise-and-fall dynamics reveal a molecular and cellular vulnerability axis in prion-like α -synuclein propagation. *bioRxiv*. doi: <https://doi.org/10.64898/2026.03.27.714785> . 2026.

Awards & Scholarships

2020–24	Aker Scholarship , competitive national scholarship for Norwegian students pursuing graduate study at leading international universities covering tuition, living costs, conferences, and travel	Oslo, Norway
2023	Linacre College Academic Travel Grant , Grant awarded for research stay at Alzheimer Center Amsterdam	Oxford, UK

CONFERENCES, TALKS & LECTURES

INVITED CONFERENCE & WORKSHOP TALKS

Feb 2026	Workshop in New Trends in Mathematics and Brain Mechanics , Talk: <i>Modeling neurodegenerative disease progression and neuronal activity</i>	Oslo, Norway
Sept 2023	Bernstein Conference , Minisymposium: “Whole-brain Dynamics: Modeling and Applications”	Berlin, DE
Apr 2023	British Applied Mathematics Conference , Minisymposium: “Neurodynamics”	Bristol, UK
Sept 2022	International Conference on Clinical Neurophysiology , Symposium: “Translational Computational Modelling Relates AD Pathophysiology to Large-Scale Brain Dynamics”	Geneva, CH

INVITED SEMINARS

Mar 2025	Oxford–GSK Computational Medicine Center Seminar , University of Oxford	Oxford, UK
Jan 2025	Network Science Institute Seminar , Northeastern University	Boston, USA
Feb 2025	Computational Neuroscience Initiative Seminar , University of Pennsylvania	Philadelphia, USA
Nov 2024	Brain Connectivity Workshop Seminar Series , University of Pennsylvania	Philadelphia, USA
Jun 2024	Department of Mathematics Seminar Series , Vrije Universiteit	Amsterdam, NL
Jan 2024	Linacre College Seminar Series , University of Oxford	Oxford, UK

INVITED GUEST LECTURE

Sep 2025	Network Neuroscience (Graduate course, instr. Dani Bassett) , University of Pennsylvania	Philadelphia, USA
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INVITED RESEARCH CENTER & GROUP TALKS

May 2025	Alzheimer’s Center , Amsterdam University Medical Center	Amsterdam, NL
Jun 2024	Network Neuroscience Group , Amsterdam Universitiy Medical Center	Amsterdam, NL
Jun 2024	Alzheimer’s Center , Amsterdam University Medical Center	Amsterdam, NL
Sept 2024	Methods Group , University College London	London, UK
Dec 2023	Center for Brain and Cognition , Pompeu Fabra University	Barcelona, ES
Nov 2021	Alzheimer’s Center , Amsterdam University Medical Center	Amsterdam, NL
Dec 2021	Mathematical Neuroscience Group , University of Nottingham	Nottingham, UK
Nov 2021	Neuromodeling Group , University of Birmingham	Birmingham, UK

CONTRIBUTED TALKS AND POSTERS

June 2026	International School and Conference on Network Science (NetSci) , Talk: “Disease spreading on brain networks”	Boston, USA
Sept 2022	Bernstein Conference , Poster: “Multi-scale Model of Alzheimer’s Disease”	Berlin, DE

ORGANIZATIONAL EXPERIENCE

Network Neuroscience Satellite Symposium, NetSci 2026 June 2026, Boston, MA, USA
Co-organized the satellite symposium “Network Neuroscience” at the International School and Conference on Network Science (NetSci 2026), coordinating speakers and program development.

Oxford Mathematical Brain Modeling Seminars Sept. 2022 - May 2024, Oxford, UK
Organized biweekly seminars on mathematical & computational neuroscience (hybrid and in-person talks).

International Conference on Mathematical Neuroscience (ICMNS 2022) July 2022, Online
Organized the minisymposium “Dynamical Systems for Neurological Disorders.”

TEACHING EXPERIENCE

2024	Python Demonstrator, <i>Computational Mathematics</i> , University of Oxford, Mathematical Institute	Oxford, UK
2023	Teaching Assistant, <i>Nonlinear Systems</i> , University of Oxford, Mathematical Institute	Oxford, UK
2022	Teaching Assistant, <i>Networks</i> , University of Oxford, Mathematical Institute	Oxford, UK
2021	Teaching Assistant, <i>Continuous Optimisation</i> , University of Oxford, Mathematical Institute	Oxford, UK
2018–2019	Teaching Assistant, <i>Mathematical Methods B</i> , NTNU, Department of Mathematical Sciences	Trondheim, Norway
2017	Teaching Laboratory Assistant, <i>Organic Chemistry</i> , NTNU, Department of Chemistry	Trondheim, Norway

SUPERVISION & MENTORING

BSc Student Mentorship at the University of Pennsylvania *Mar 2026–Present, Philadelphia, USA*
Equation discovery for neurodegenerative disease progression. Co-mentor: Prof. Dani S. Bassett.

Phd Student Mentorship at the University of Pennsylvania *July 2025–Present, Philadelphia, USA*
Control of phase-oscillator networks. Co-mentor: Prof. Dani S. Bassett.

MSc Student Mentorship at the University of Pennsylvania *March 2025–Present, Philadelphia, USA*
Computational model for predicting disease initiation in Alzheimer's. Co-mentor: Prof. Dani S. Bassett.

MSc Student Mentorship at the University of Amsterdam *Sept 2024–July 2025, Amsterdam, Netherlands*
Computational modeling of working memory deficits in Alzheimer's disease. Co-mentors: Profs. Jorge Mejias & Christian Bick.

OUTREACH

The Network Pages *June 2022, Online*
Article on modeling neural oscillations titled "Oscillators and Alzheimer's"
<https://www.networkpages.nl/oscillators-and-alzheimers/>